



PMU Swimming Pool Reservation App

Software Project Management Documentation

Course: Software Project Management SOEN 4313

Project: PMU Swimming Pool Reservation App

University: Prince Mohammad Bin Fahd University

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1. Introduction

1.1 Overview

The PMU Swimming Pool Reservation App is a mobile application idea designed to make swimming pool reservations easier for students at Prince Mohammad Bin Fahd University. Instead of depending on manual booking or asking about available times, students can use the app to check the swimming pool schedule and reserve a suitable time slot.

The system includes three main types of users: students, the pool manager, and the system administrator. Students use the app to view availability and make reservations. The pool manager controls reservations and updates available times. The system administrator manages users and system access.

The main goal of this project is to organize the reservation process, reduce booking conflicts, and make the swimming pool easier to manage.

This project follows the course project requirements, which ask for a project management document, a short PowerPoint presentation, and a demo using a PM tool such as Jira, Asana, or Redmine. The document also includes scoping, planning, launching, monitoring, closing, and PM tool screenshots.

1.2 Problem

The current swimming pool reservation process can become confusing if it is managed manually. Students may not know which time slots are available, and the pool manager may need to handle many requests at the same time. This can lead to double bookings, delays, and miscommunication.

For example, two students might request the same time slot, or a student might not receive a clear confirmation. These small issues can affect the overall experience and make the process harder for both students and the pool manager.

The proposed mobile app solves this problem by showing available times clearly and allowing students to reserve online. It also helps the pool manager organize reservations in one place.

1.3 Deliverables

The main deliverables of this project are:

1. A complete project management document.

2. A short PowerPoint presentation.
3. A Jira project management tool demo.
4. Functional and non-functional requirements.
5. Requirement Breakdown Structure.
6. Project Overview Statement.
7. Work Breakdown Structure.
8. Detailed task list.
9. Task duration estimation.
10. Resource requirement estimation.
11. Cost analysis and total project cost.
12. Project management life cycle model.
13. Task-on-Node network.
14. Gantt chart.
15. Risk plan.
16. Scope change request rules and form.
17. Responsibility matrix.
18. Decision-making rules.
19. Communication rules.
20. Monitoring schedule and budget.
21. Earned Value Analysis.
22. S-Curve.
23. Post-Implementation Audit Report.
24. Final deliverables list.
25. Jira screenshots.
26. Group meeting minutes.

2. Scoping

2.1 System Boundaries

System boundaries explain what is included inside the system and what is not included. This helps the team understand the project scope clearly and avoid adding unnecessary features.

Inside the System Scope

The system will include:

- Student login and account access.
- Viewing swimming pool availability.
- Reserving available swimming pool time slots.
- Canceling reservations.
- Pool manager reservation control.
- Pool manager availability updates.
- Admin user management.
- Reservation confirmation notifications.
- Basic reservation history.

Outside the System Scope

The system will not include:

- Online payment system.
- Fitness tracking.
- Swimming training programs.
- Smart card access to the swimming pool.
- Integration with university attendance systems.
- Advanced analytics dashboard.
- Biometric authentication.

2.2 Conditions of Satisfaction

The project will be considered successful if the following conditions are achieved:

1. Students can view available swimming pool time slots.
2. Students can reserve an available time slot.
3. Students can cancel their reservations.
4. The pool manager can approve, reject, and manage reservations.
5. The pool manager can update available and unavailable times.
6. The system admin can manage users and roles.
7. The system prevents double booking.
8. The system is simple and easy to use.
9. The project document includes all required Software Project Management sections.
10. The team provides Jira screenshots as evidence of PM tool usage.
11. The PowerPoint presentation clearly explains the project.

2.3 Functional Requirements

No.	Functional Requirement
FR1	Students shall be able to create an account and log in to the mobile application.
FR2	Students shall be able to view swimming pool availability by date and time.
FR3	Students shall be able to reserve an available swimming pool time slot.
FR4	Students shall be able to cancel their reservation before the scheduled time.
FR5	The pool manager shall be able to approve, reject, or manage reservations.
FR6	The pool manager shall be able to update available and unavailable pool time slots.
FR7	The system admin shall be able to manage user accounts and roles.
FR8	The system shall send reservation confirmation notifications to students.

2.4 Non-Functional Requirements

No.	Non-Functional Requirement
NF R1	The application shall be user-friendly and easy to use for students, pool managers, and admins.
NF R2	The system shall protect user data using secure login and access control.
NF R3	The application shall respond to user actions within 3 seconds under normal usage.
NF R4	The system shall be available during university working hours with minimum downtime.

2.5 Requirement Breakdown Structure

The Requirement Breakdown Structure divides the system requirements into smaller groups. This makes the requirements easier to understand and manage.

1. Student Requirements

1.1 Student login

1.2 View pool availability

1.3 Reserve time slot

1.4 Cancel reservation

1.5 View reservation history

2. Pool Manager Requirements

2.1 View all reservations

2.2 Approve reservations

2.3 Reject reservations

2.4 Update pool availability

2.5 Manage unavailable time slots

3. Admin Requirements

- 3.1 Manage user accounts
- 3.2 Assign user roles
- 3.3 Manage system access
- 3.4 Maintain system settings

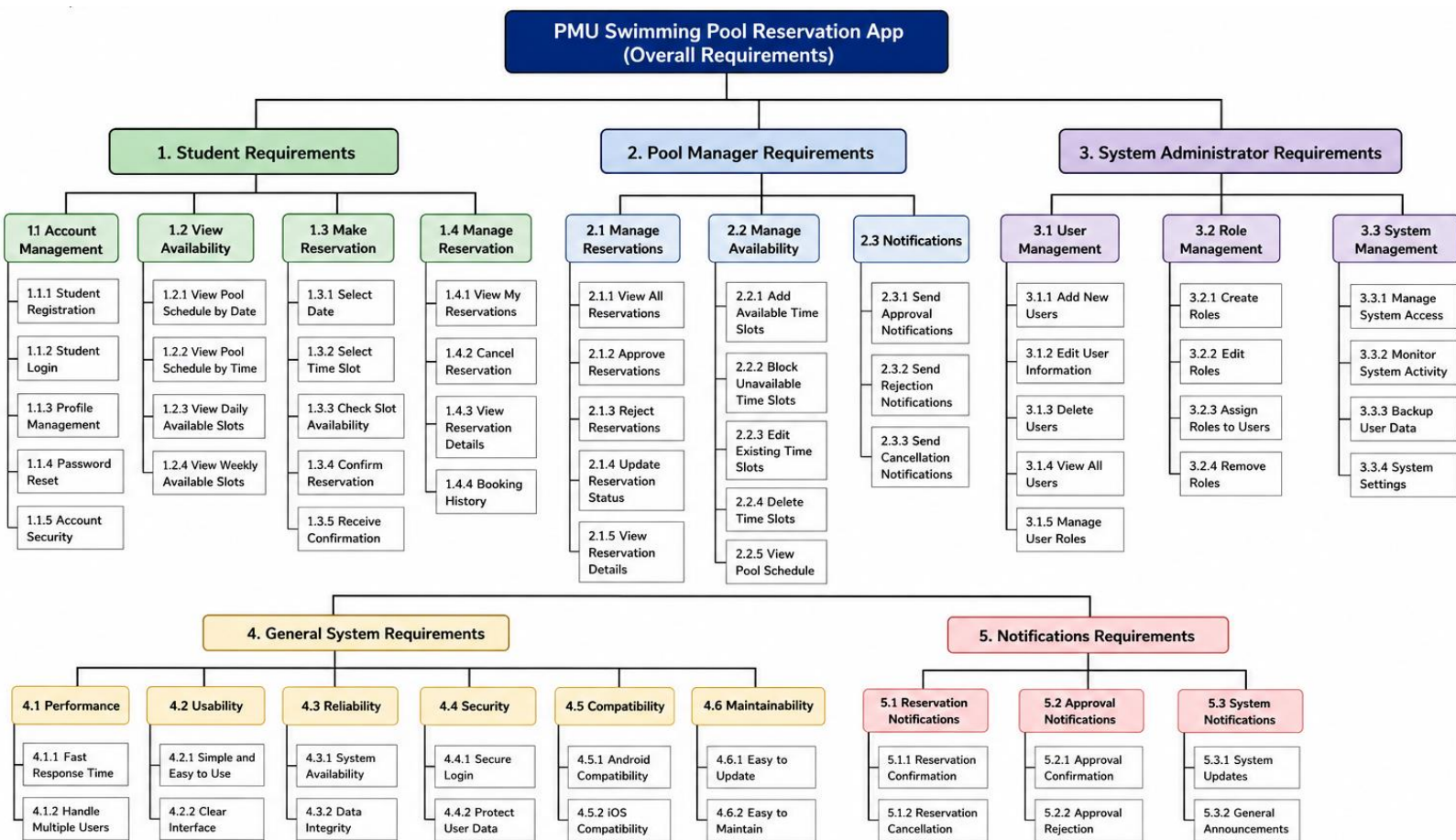
4. Notification Requirements

- 4.1 Send reservation confirmation
- 4.2 Send cancellation confirmation
- 4.3 Notify students of reservation status

5. Security Requirements

- 5.1 Secure login
- 5.2 Role-based access control
- 5.3 Data protection

2.5.1 Requirement Breakdown Structure



2.6 Project Overview Statement

Item	Description
Project Name	PMU Swimming Pool Reservation App
Project Goal	To create a mobile application concept that allows PMU students to reserve the swimming pool online.
Problem	Manual reservation methods can cause confusion, double booking, and poor communication.

Objectives	Improve reservation management, reduce booking conflicts, allow students to view availability, and help the pool manager control bookings.
Success Criteria	The app supports student reservations, manager control, admin management, and prevents booking conflicts.
Assumptions	Students have mobile devices, the university supports online reservation, and the pool manager will update availability.
Risks	Requirement changes, time limitations, missing data, and team coordination issues.
Constraints	Limited project time, academic scope, and no real payment or university system integration.

3. Planning

3.1 Work Breakdown Structure

The Work Breakdown Structure divides the project into smaller work packages. This helps the team organize the work and assign responsibilities clearly.

Level 1: PMU Swimming Pool Reservation App

1. Project Initiation

1.1 Define project idea

1.2 Identify users

1.3 Write overview

1.4 Define problem

2. Scoping

2.1 Define system boundaries

2.2 Write conditions of satisfaction

- 2.3 Create functional requirements
- 2.4 Create non-functional requirements
- 2.5 Create RBS
- 2.6 Create POS

3. Planning

- 3.1 Create WBS
- 3.2 Prepare detailed task list
- 3.3 Estimate task durations
- 3.4 Estimate resources
- 3.5 Prepare cost analysis
- 3.6 Create Gantt chart
- 3.7 Create Task-on-Node network
- 3.8 Prepare risk plan

4. Launching

- 4.1 Create scope change request rules
- 4.2 Create responsibility matrix
- 4.3 Define decision-making rules
- 4.4 Define communication rules

5. Monitoring

- 5.1 Monitor schedule
- 5.2 Monitor budget
- 5.3 Create Earned Value Analysis
- 5.4 Create S-Curve

6. Closing

- 6.1 Prepare Post-Implementation Audit Report
- 6.2 List final deliverables
- 6.3 Review final document
- 6.4 Prepare presentation

7. PM Tool

7.1 Create Jira project

7.2 Add tasks

7.3 Assign tasks

7.4 Add meeting minutes

7.5 Take screenshots

3.2 Detailed Task List

Task Name	Description	Assignee	Priority	Due Date
Write project introduction	Write the project overview and introduce the main users.	Rakan Alshalwi	Medium	Mar 5, 2026
Define project problem	Explain the issue with manual swimming pool reservations.	Rakan Alshalwi	Medium	Mar 5, 2026
Create cost analysis	Estimate human resources, tools, and project budget.	Rakan Alshalwi	High	Mar 20, 2026
Create Earned Value Analysis	Prepare EVM table using PV, EV, AC, CV, SV, CPI, and SPI.	Rakan Alshalwi	High	Apr 13, 2026
Final document review	Review formatting, sections, and final project quality.	Rakan Alshalwi	High	May 5, 2026
Prepare Jira demo screenshots	Take screenshots of the Jira board, tasks, and meeting minutes.	Rakan Alshalwi	High	May 5, 2026
Create functional requirements	Write at least six functional requirements.	Abdulrahman Ali Albaz	High	Mar 7, 2026
Create non-functional requirements	Write at least three non-functional requirements.	Abdulrahman Ali Albaz	High	Mar 7, 2026

Create Gantt chart	Create project timeline with start and end dates.	Abdulrahman Ali Albaz	High	Mar 27, 2026
Create PowerPoint presentation	Prepare project presentation slides.	Abdulrahman Ali Albaz	High	May 5, 2026
Create S-Curve	Show planned and actual progress over time.	Abdulrahman Ali Albaz	High	May 4, 2026
Define system boundaries	Define what is included and excluded from the system.	Abdulaziz Hashim	Medium	Mar 8, 2026
Create RBS	Create Requirement Breakdown Structure.	Abdulaziz Hashim	High	Mar 10, 2026
Create POS	Create Project Overview Statement.	Abdulaziz Hashim	High	Mar 11, 2026
Create Task-on-Node network	Create diagram showing task dependencies.	Abdulaziz Hashim	High	Mar 23, 2026
Create WBS	Divide the project into clear work packages.	Naif Nasser Alharbi	High	Mar 14, 2026
Estimate task durations	Estimate duration for each project task.	Naif Nasser Alharbi	High	Mar 17, 2026
Create risk plan	Identify project risks and response strategies.	Naif Nasser Alharbi	High	Mar 29, 2026
Create responsibility matrix	Assign responsibilities to team members.	Naif Nasser Alharbi	High	Mar 25, 2026
Prepare detailed task list	Prepare a complete task list based on the project sections.	Nahidh	High	Mar 15, 2026

Estimate resource requirements	Identify team, software, and tool resources.	Nahidh	Medium	Mar 18, 2026
Write communication rules	Define team communication methods and rules.	Nahidh	Medium	Mar 25, 2026
Write decision-making rules	Define how decisions and disagreements are handled.	Nahidh	Medium	Mar 26, 2026
Group Meeting Minutes 1	Record first group meeting notes.	Rakan Alshalwi	Medium	Mar 3, 2026
Group Meeting Minutes 2	Record second group meeting notes.	Nahidh	Medium	Mar 17, 2026

3.3 Estimating Task Duration

Task	Duration
Write project introduction	2 days
Define project problem	2 days
Create functional requirements	3 days
Create non-functional requirements	3 days
Define system boundaries	2 days
Create RBS	4 days
Create POS	3 days
Create WBS	3 days
Prepare detailed task list	3 days
Estimate task durations	3 days
Estimate resource requirements	3 days
Create cost analysis	2 days

Create Gantt chart	4 days
Create Task-on-Node network	4 days
Create responsibility matrix	2 days
Write communication rules	1 day
Write decision-making rules	2 days
Create risk plan	3 days
Create Earned Value Analysis	3 days
Create S-Curve	3 days
Create PowerPoint presentation	3 days
Prepare Jira screenshots	2 days
Final document review	2 days

3.4 Estimating Resource Requirements

Resource	Purpose
Team members	Complete project documentation, planning, and presentation.
Laptop/PC	Prepare the document, presentation, and Jira project.
Microsoft Word	Write the project management document.
Microsoft PowerPoint	Create the final presentation.
Jira	Manage tasks, assignments, Kanban board, and meeting minutes.
Internet connection	Access Jira and share project files.
Diagram tool	Create WBS, RBS, Gantt chart, and Task-on-Node network.
Excel	Prepare cost analysis, EVM, and S-Curve.

3.5 Cost Analysis and Project Cost

This is an academic project, so the cost is estimated based on student work effort and available tools. Most tools used for the project are free or already available to the team.

Human Resource Cost

Assumption: Each team member work hour is estimated at 50 SAR.

Team Member	Estimated Hours	Rate per Hour	Cost
Rakan Alshalwi	14	50 SAR	700 SAR
Abdulrahman Ali Albaz	10	50 SAR	500 SAR
Abdulaziz Hashim	10	50 SAR	500 SAR
Naif Nasser Alharbi	10	50 SAR	500 SAR
Nahidh	8	50 SAR	400 SAR
Total	52 hours		2600 SAR

Tool Cost

Tool	Cost
Jira free plan	0 SAR
Microsoft Word	0 SAR
Microsoft PowerPoint	0 SAR
Excel	0 SAR
Internet and personal laptops	0 SAR

Total Project Cost

Cost Type	Amount
Human resource cost	2600 SAR

Tool cost	0 SAR
Other cost	0 SAR
Total Estimated Cost	2600 SAR

3.6 Project Management Life Cycle Model

The project follows a simple predictive project management life cycle because the project requirements are clear, and the deliverables are known from the beginning.

The project phases are:

1. Initiation

The team understands the project idea, users, and main problem.

2. Scoping

The team defines system boundaries, requirements, RBS, and POS.

3. Planning

The team creates the WBS, task list, cost analysis, schedule, and risk plan.

4. Launching

The team defines responsibilities, communication rules, decision-making rules, and change request rules.

5. Monitoring

The team uses Jira, EVM, and S-Curve to monitor project progress.

6. Closing

The team prepares the final deliverables, presentation, and audit report.

3.7 Schedule

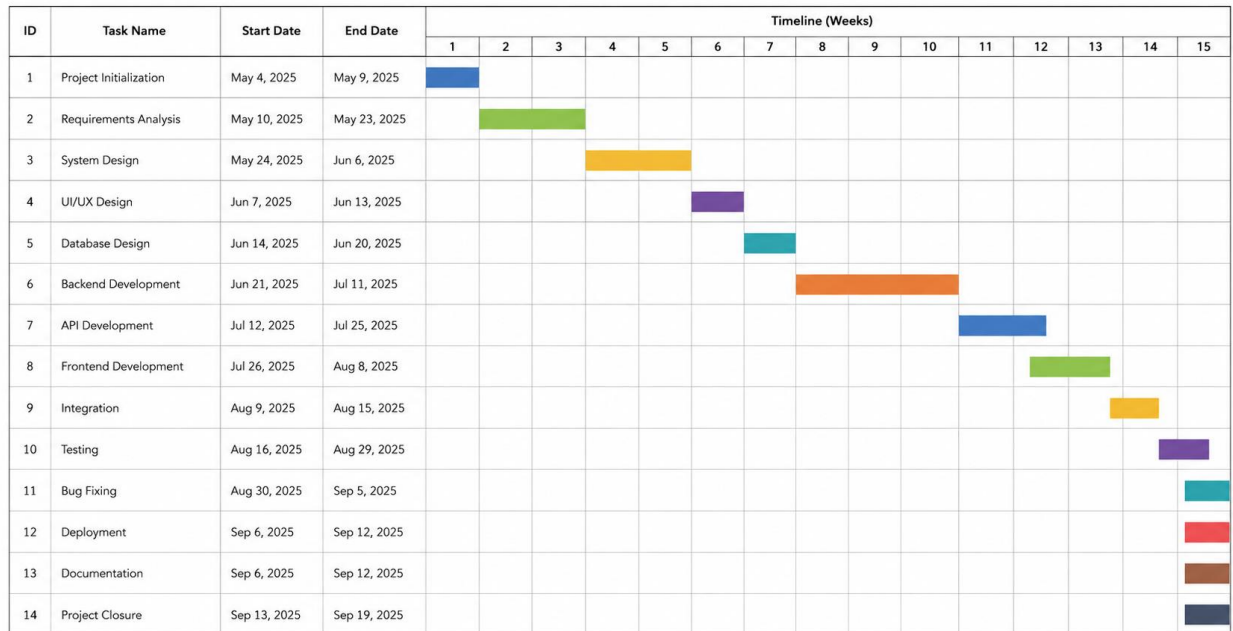
3.7.1 Gantt Chart Table

Task	Start Date	End Date
Project introduction	Mar 3, 2026	Mar 5, 2026
Project problem	Mar 3, 2026	Mar 5, 2026

Functional requirements	Mar 4, 2026	Mar 7, 2026
Non-functional requirements	Mar 4, 2026	Mar 7, 2026
System boundaries	Mar 6, 2026	Mar 8, 2026
RBS	Mar 6, 2026	Mar 10, 2026
POS	Mar 8, 2026	Mar 11, 2026
WBS	Mar 11, 2026	Mar 14, 2026
Detailed task list	Mar 12, 2026	Mar 15, 2026
Task duration estimation	Mar 14, 2026	Mar 17, 2026
Resource estimation	Mar 15, 2026	Mar 18, 2026
Cost analysis	Mar 18, 2026	Mar 20, 2026
Gantt chart	Mar 19, 2026	Mar 27, 2026
Task-on-Node network	Mar 19, 2026	Mar 23, 2026
Responsibility matrix	Mar 23, 2026	Mar 25, 2026
Communication rules	Mar 24, 2026	Mar 25, 2026
Decision-making rules	Mar 24, 2026	Mar 26, 2026
Risk plan	Mar 26, 2026	Mar 29, 2026
Earned Value Analysis	Mar 30, 2026	Apr 13, 2026
S-Curve	Apr 1, 2026	May 4, 2026
PowerPoint presentation	Apr 8, 2026	May 5, 2026
Jira screenshots	Apr 10, 2026	May 5, 2026
Final document review	Apr 12, 2026	May 5, 2026

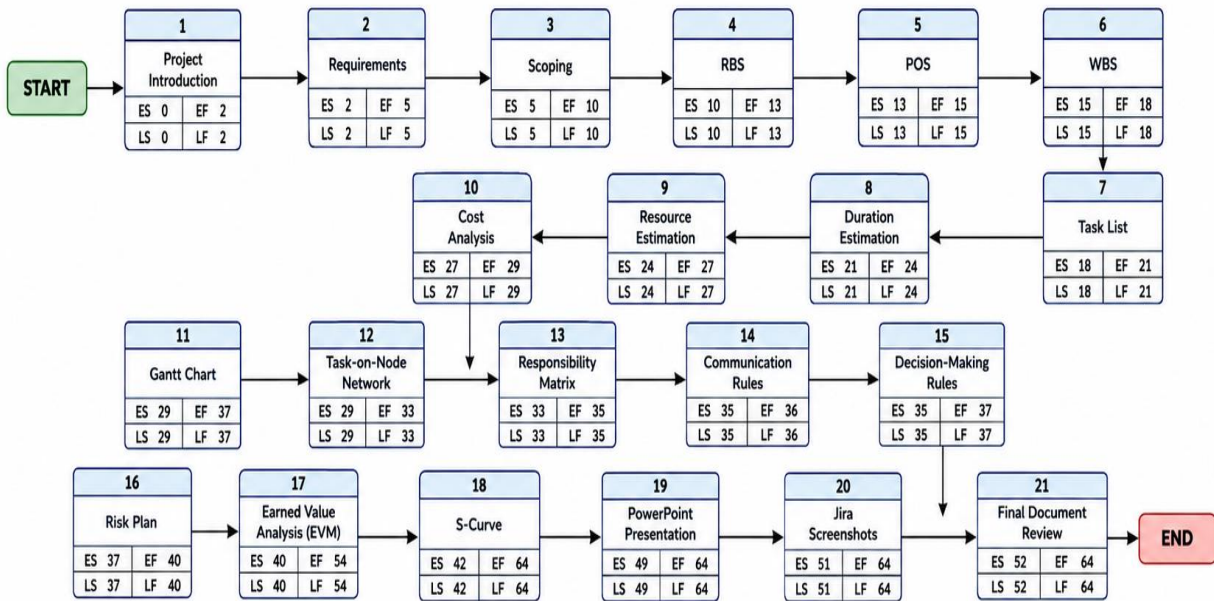
3.7.2 Gantt Chart Diagram

PMU Swimming Pool Reservation App – Gantt Chart



3.8 Task-on-Node Network

The Task-on-Node network shows the order of the main project tasks and how they depend on each other.



Task Dependencies

Code	Task	Predecessor
A	Project introduction	None
B	Problem statement	A
C	Functional requirements	B
D	Non-functional requirements	B
E	System boundaries	C, D
F	RBS	E
G	POS	F
H	WBS	G
I	Detailed task list	H
J	Duration estimation	I
K	Resource estimation	I
L	Cost analysis	J, K
M	Gantt chart	J

N	Task-on-Node network	J
O	Responsibility matrix	H
P	Communication rules	O
Q	Decision-making rules	O
R	Risk plan	L, M
S	EVM	M, L
T	S-Curve	S
U	PowerPoint presentation	All previous planning tasks
V	Jira screenshots	Jira tasks completed
W	Final document review	U, V, T

Network Flow

Start → A → B → C/D → E → F → G → H → I → J/K → L/M/N → O/P/Q → R → S → T → U/V → W → Finish

3.9 Risk Plan

Risk	Probability	Impact	Response Plan
Requirement misunderstanding	Medium	High	Review the project description carefully and confirm requirements with the team.
Team member delay	Medium	Medium	Assign backup members and monitor tasks using Jira.
Missing documentation section	Low	High	Use a checklist based on the project instructions.
Jira setup issue	Medium	Medium	Use available screenshots and prepare backup documentation manually.

Poor time estimation	Medium	Medium	Review task durations and update the schedule when needed.
Scope creep	Medium	High	Use a scope change request form before accepting any change.
Presentation not ready	Low	High	Prepare slides before the final deadline and rehearse the demo.
Weak communication	Medium	Medium	Use clear communication rules and meeting minutes.

4. Launching

4.1 Scope Change Request Rules

To keep the project organized, the team will follow clear rules for any scope change.

1. Any proposed change must be written in a scope change request form.
2. The change must include the reason, impact, cost, and schedule effect.
3. The team must review the change before approving it.
4. Major changes must be approved by majority decision.
5. Approved changes will be added to the project plan.
6. Rejected changes will not be included in the project scope.

4.2 Scope Change Request Form

Field	Description
Change Request ID	SCR-001
Request Date	March 25, 2026

Requested By	Team Member
Change Description	Describe the requested change.
Reason for Change	Explain why the change is needed.
Impact on Scope	Explain how the project scope will change.
Impact on Schedule	Explain if more time is needed.
Impact on Cost	Explain if cost will increase.
Decision	Approved / Rejected
Approved By	Team Leader / Majority Decision

4.3 Responsibility Matrix

Legend:

R = Responsible

C = Consulted

Task	Rakan	Abdulrahman	Abdulaziz	Naif	Nahidh
Introduction	R	C	C	C	C
Problem statement	R	C	C	C	C
Functional requirements	C	R	C	C	C
Non-functional requirements	C	R	C	C	C
System boundaries	C	C	R	C	C
RBS	C	C	R	C	C
POS	C	C	R	C	C
WBS	C	C	C	R	C
Detailed task list	C	C	C	C	R
Duration estimation	C	C	C	R	C

Resource estimation	C	C	C	C	R
Cost analysis	R	C	C	C	C
Gantt chart	C	R	C	C	C
Task-on-Node network	C	C	R	C	C
Risk plan	C	C	C	R	C
Responsibility matrix	C	C	C	R	C
Communication rules	C	C	C	C	R
Decision-making rules	C	C	C	C	R
EVM	R	C	C	C	C
S-Curve	C	R	C	C	C
PowerPoint	C	R	C	C	C
Jira screenshots	R	C	C	C	C
Final review	R	C	C	C	C

4.4 Decision-Making Rules

The team will follow these decision-making rules:

1. Simple project decisions are made through team discussion.
2. Important decisions are made by majority vote.
3. If there is disagreement, the team will compare the options based on time, quality, and project requirements.
4. If the vote is equal, the team leader will make the final decision.
5. All major decisions will be recorded in meeting minutes.
6. Approved decisions must be followed by all team members.

4.5 Communication Rules

The team will follow these communication rules:

1. Team communication will be done through WhatsApp and in-person meetings.
2. Jira will be used to track tasks and responsibilities.

3. The team will hold meetings when important sections need review.
4. Each member should update task progress regularly.
5. Important decisions must be written in the meeting minutes.
6. Team members should reply to project messages within 24 hours.
7. Final documents and screenshots will be shared with all members before submission.

5. Monitoring

5.1 Monitoring Schedule and Budget

The team used Jira to monitor the project progress. The Kanban board was divided into four columns:

1. To Do
2. In Progress
3. In Review
4. Done

Each task was assigned to a team member with a due date and priority. This made it easier to see which tasks were completed, which tasks were still being worked on, and which tasks needed review.

The budget was monitored using planned value, earned value, and actual cost. Since this is an academic project, most costs are based on estimated student work hours.

5.2 Earned Value Analysis

EVM Assumptions

- Total planned project budget: 2600 SAR
- Planned project completion at monitoring point: 80%
- Actual project completion at monitoring point: 75%
- Actual cost at monitoring point: 2100 SAR

EVM Calculations

Metric	Formula	Value
Budget at Completion BAC	Total project budget	2600 SAR
Planned Value PV	$BAC \times \text{Planned \%}$	$2600 \times 0.80 = 2080 \text{ SAR}$
Earned Value EV	$BAC \times \text{Actual \%}$	$2600 \times 0.75 = 1950 \text{ SAR}$
Actual Cost AC	Actual cost spent	2100 SAR
Cost Variance CV	$EV - AC$	$1950 - 2100 = -150 \text{ SAR}$
Schedule Variance SV	$EV - PV$	$1950 - 2080 = -130 \text{ SAR}$
Cost Performance Index CPI	EV / AC	$1950 / 2100 = 0.93$
Schedule Performance Index SPI	EV / PV	$1950 / 2080 = 0.94$

EVM Interpretation

The project has a negative cost variance of -150 SAR, which means the project is slightly over budget. The schedule variance is -130 SAR, which means the project is slightly behind schedule. The CPI is 0.93, so the cost performance is slightly below the planned level. The SPI is 0.94, so the project is progressing a little slower than planned.

However, the differences are small and can be controlled by completing the remaining tasks before the final deadline.

5.3 S-Curve

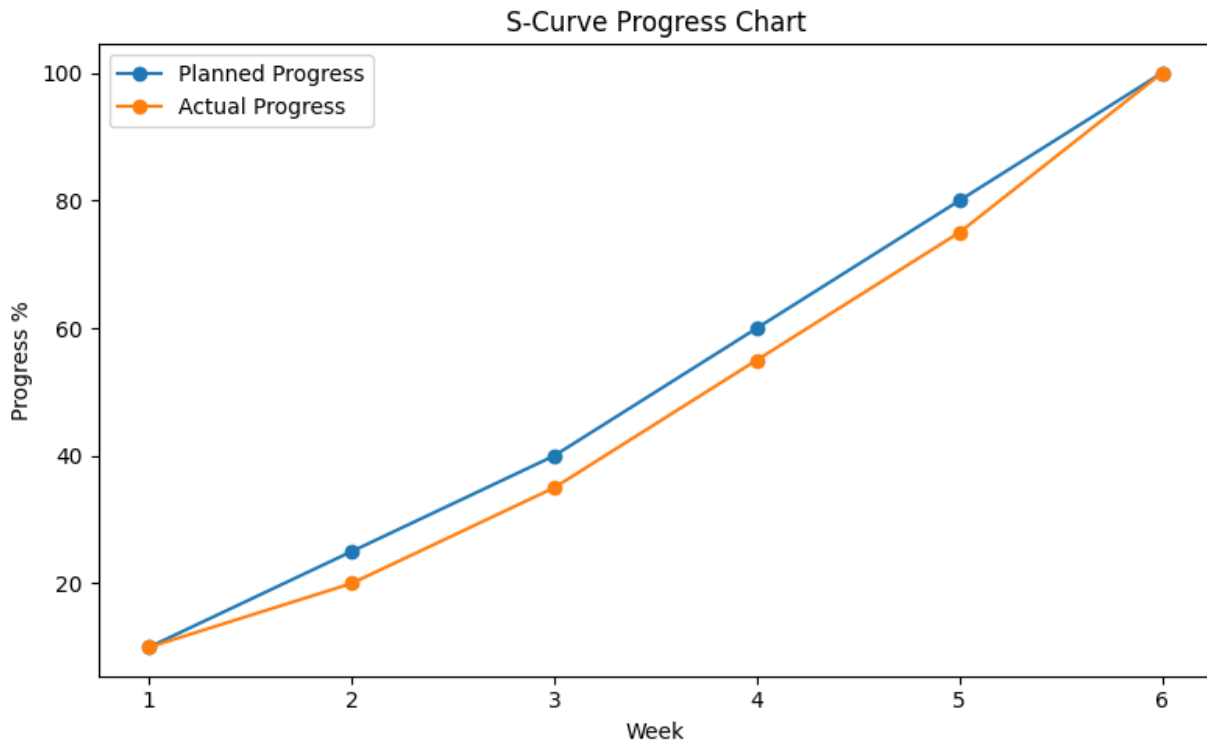
The S-Curve shows planned progress compared to actual progress over time.

Week	Planned Progress	Actual Progress
Week 1	10%	10%
Week 2	25%	20%

Week 3	40%	35%
Week 4	60%	55%
Week 5	80%	75%
Week 6	100%	100%

S-Curve Interpretation

The actual progress was slightly lower than the planned progress during the middle of the project. This happened because some planning and monitoring tasks took more time than expected. By the final week, the team completed the remaining tasks and reached 100% completion.



Visual S-Curve

6. Closing

6.1 Post-Implementation Audit Report

Project Name

PMU Swimming Pool Reservation App

Project Objective

The objective of this project was to create a complete Software Project Management document for a mobile application that manages online swimming pool reservations at PMU.

Project Results

The project achieved its main objectives. The team prepared the required project management sections, including introduction, scoping, planning, launching, monitoring, closing, and PM tool screenshots. Jira was used to manage tasks, assign responsibilities, and monitor progress.

What Went Well

1. The project scope was clear from the beginning.
2. Team responsibilities were divided clearly.
3. Jira helped the team track progress.
4. Meeting minutes helped document team decisions.
5. The project document covered all required sections.

What Could Be Improved

1. More time could be given to creating diagrams.
2. Task due dates could be updated more regularly.
3. The team could add more detailed screenshots from Jira.
4. The app prototype could be expanded in the future.

Lessons Learned

1. Project planning helps organize work and reduce confusion.
2. Clear task assignment improves team coordination.

3. Jira is useful for tracking task progress.
4. Risk planning helps the team prepare for possible problems.
5. Monitoring tools such as EVM and S-Curve help evaluate project performance.

6.2 Final Deliverables

The final project deliverables are:

1. Software Project Management document.
2. Functional requirements.
3. Non-functional requirements.
4. System boundaries.
5. Conditions of satisfaction.
6. Requirement Breakdown Structure.
7. Project Overview Statement.
8. Work Breakdown Structure.
9. Detailed task list.
10. Task duration estimation.
11. Resource requirement estimation.
12. Cost analysis.
13. Gantt chart.
14. Task-on-Node network.
15. Risk plan.
16. Scope change request form.
17. Responsibility matrix.
18. Communication rules.
19. Decision-making rules.
20. Earned Value Analysis.
21. S-Curve.
22. Post-Implementation Audit Report.
23. Jira screenshots.

- 24. Group meeting minutes.
- 25. PowerPoint presentation.

7. PM Tool Screenshots

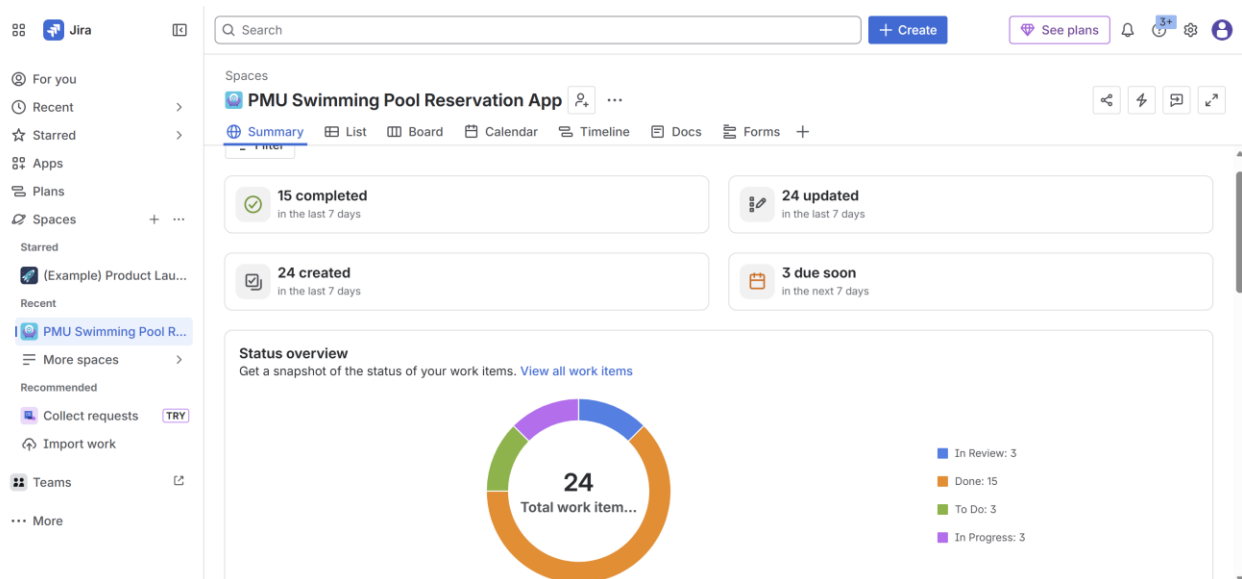
7.1 PM Tool Used

The team used Jira to organize and track the project work. A Kanban board was created for the PMU Swimming Pool Reservation App, and the tasks were divided into four columns: To Do, In Progress, In Review, and Done.

Each task was added to Jira with a clear description, assignee, priority, and due date. This made it easier to see who was responsible for each part of the project and what still needed to be completed.

Jira was also used to record group meeting minutes. This helped the team keep track of important discussions, decisions, and action items. Overall, Jira made the project easier to manage and helped show the project progress in a clear way.

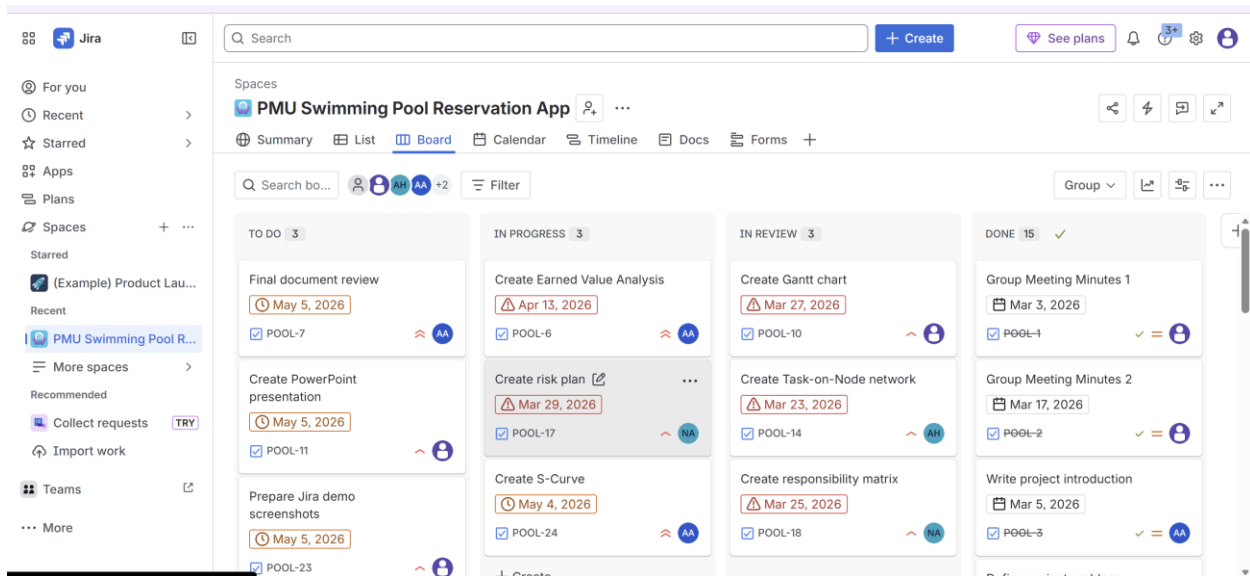
7.2 Screenshot 1: Jira Project Summary Page



Caption:

Figure 1 shows the Jira project summary page for the PMU Swimming Pool Reservation App. The page displays the total number of created, updated, completed, and due tasks.

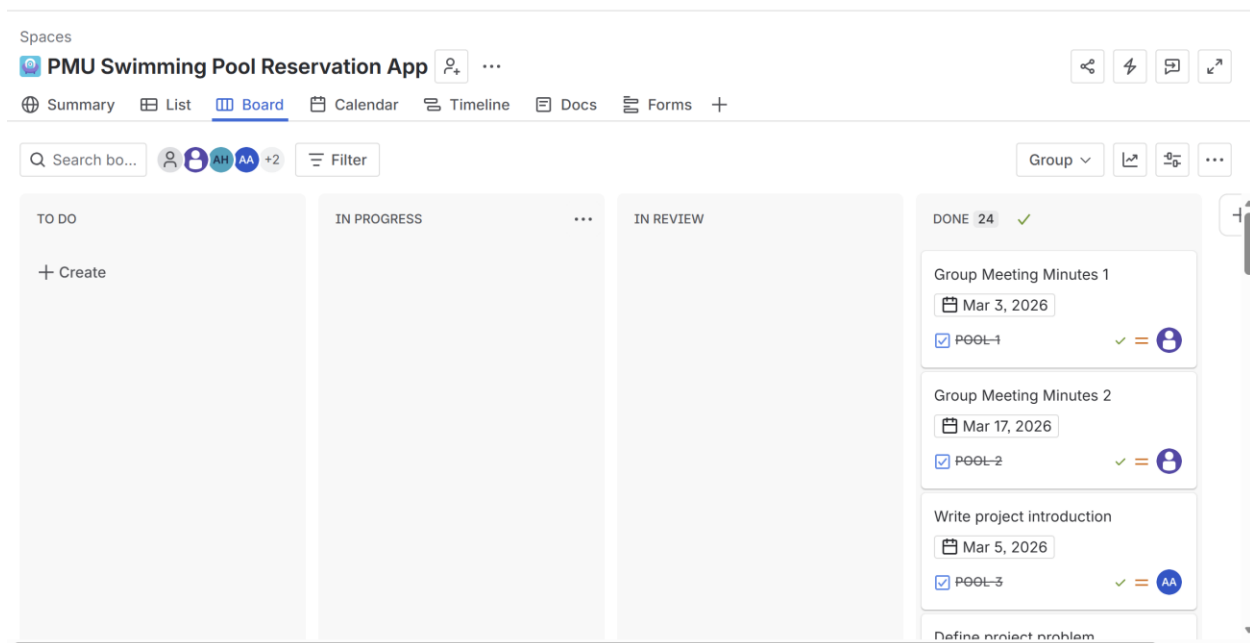
7.3 Screenshot 2: Project Progress Kanban Board



Caption:

Figure 2 shows the project Kanban board during the monitoring stage. Tasks are divided into To Do, In Progress, In Review, and Done to track the project progress.

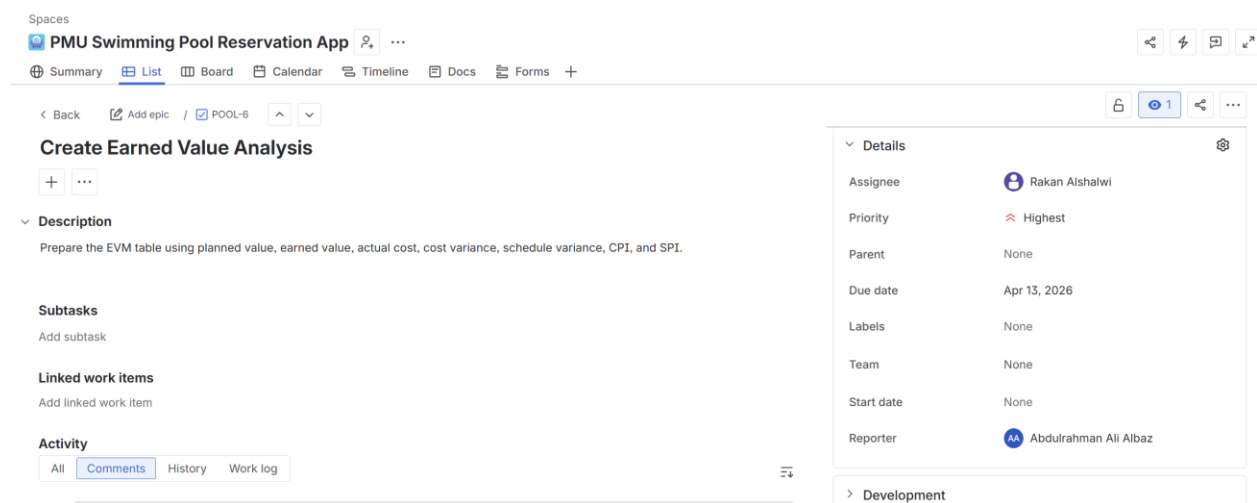
7.4 Screenshot 3: Final Completed Kanban Board



Caption:

Figure 3 shows the final Kanban board after all project tasks were completed and moved to the Done column.

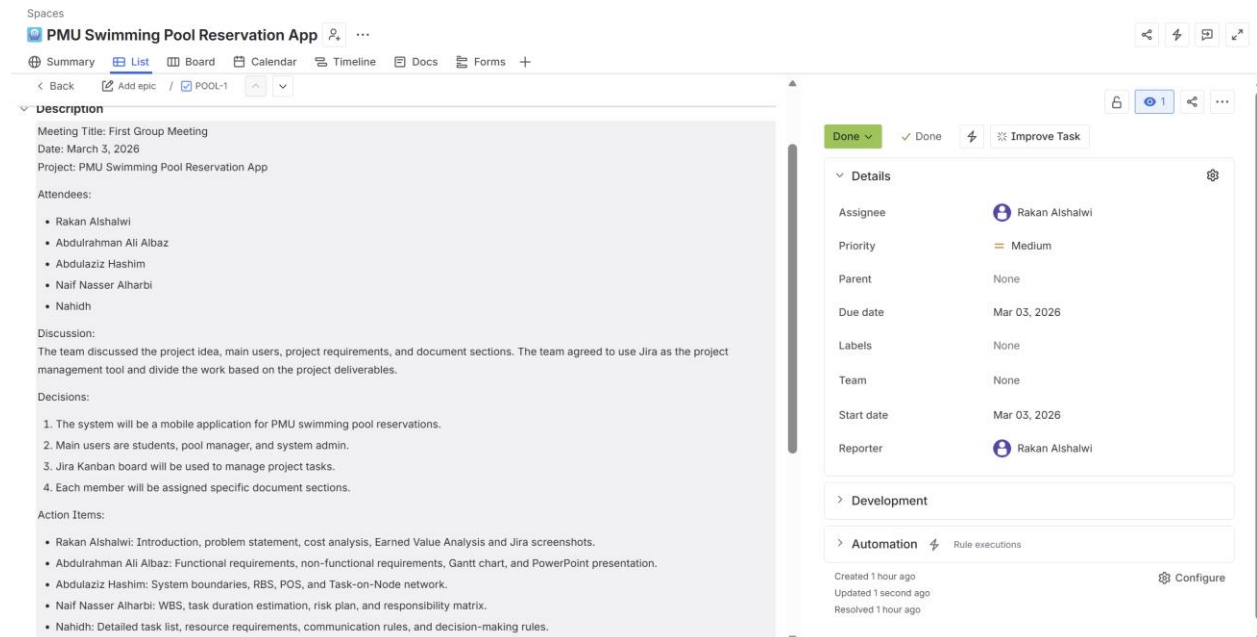
7.5 Screenshot 4: Sample Task Details



Caption:

Figure 4 shows a sample Jira task for Earned Value Analysis. The task includes a description, assignee, priority, due date, and status.

7.6 Screenshot 5: Group Meeting Minutes



Caption:

Figure 5 shows the first group meeting minutes recorded in Jira. It includes the meeting date, attendees, discussion points, decisions, and action items.

8. Appendix: Group Meeting Minutes

8.1 Group Meeting Minutes 1

Meeting Title: First Group Meeting

Date: March 3, 2026

Project: PMU Swimming Pool Reservation App

Attendees

- Rakan Alshalwi
- Abdulrahman Ali Albaz
- Abdulaziz Hashim
- Naif Nasser Alharbi
- Nahidh

Discussion

The team discussed the project idea, main users, project requirements, and document sections. The team agreed to use Jira as the project management tool and divide the work based on the project deliverables.

Decisions

1. The system will be a mobile application for PMU swimming pool reservations.
2. Main users are students, pool manager, and system admin.
3. Jira Kanban board will be used to manage project tasks.
4. Each member will be assigned specific document sections.

Action Items

- Rakan Alshalwi: Introduction, problem statement, cost analysis, Earned Value Analysis, Jira screenshots, and final document review.

- Abdulrahman Ali Albaz: Functional requirements, non-functional requirements, Gantt chart, S-Curve, and PowerPoint presentation.
- Abdulaziz Hashim: System boundaries, RBS, POS, and Task-on-Node network.
- Naif Nasser Alharbi: WBS, task duration estimation, risk plan, and responsibility matrix.
- Nahidh: Detailed task list, resource requirements, communication rules, and decision-making rules.

8.2 Group Meeting Minutes 2

Meeting Title: Second Group Meeting

Date: March 17, 2026

Project: PMU Swimming Pool Reservation App

Attendees

- Rakan Alshalwi
- Abdulrahman Ali Albaz
- Abdulaziz Hashim
- Naif Nasser Alharbi
- Nahidh

Discussion

The team reviewed the completed introduction, requirements, and scoping sections. The team also checked the Kanban board progress and moved completed tasks to the Done column.

Decisions

1. The WBS will be divided into planning, design, development, testing, documentation, and presentation.
2. The team will prepare screenshots from Jira for the final document.
3. The final document will include Kanban board screenshots and task assignments.
4. The team will review the final document before submission.

Action Items

- Rakan Alshalwi: Review introduction, deliverables, cost analysis, EVM, Jira screenshots, and final document review.
- Abdulrahman Ali Albaz: Finalize requirements, Gantt chart, S-Curve, and presentation slides.
- Abdulaziz Hashim: Finalize scoping, RBS, POS, and Task-on-Node network.
- Naif Nasser Alharbi: Finalize WBS, risk plan, and responsibility matrix.
- Nahidh: Finalize resource requirements, communication rules, and decision-making rules.

9. Conclusion

In conclusion, the PMU Swimming Pool Reservation App is a useful project idea that helps students reserve the university swimming pool in a simple and organized way. The app reduces confusion by allowing students to check available time slots and make reservations online.

The project also helps the pool manager control reservations and update availability, while the system administrator manages users and access. From a project management view, the team planned the project using WBS, RBS, POS, cost analysis, Gantt chart, Task-on-Node network, risk plan, EVM, and S-Curve.

Jira was used as the project management tool to assign tasks, monitor progress, record meeting minutes, and show the final project completion. This made the project more organized and helped the team complete the required deliverables.